Volatility Forecasting and Option Pricing Using a GARCH-LSTM Model

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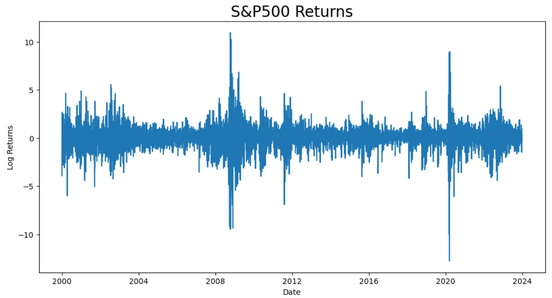
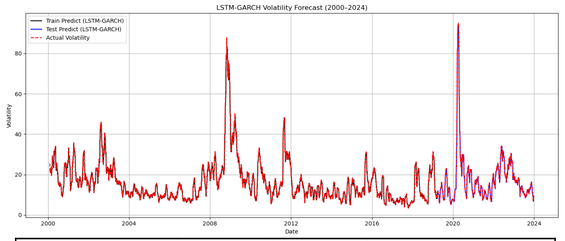
Overview

- **Volatility** plays a central role in financial markets, especially for pricing derivatives and managing risk.
- Traditional models like GARCH are effective in modeling time-varying volatility because they capture **volatility clustering.**
- This project builds a **hybrid model** that pairs **GARCH** (uses past volatility patterns) with **LSTM** (learns nonlinear time dynamics) to produce more accurate volatility forecasts.
- We also incorporate **regime-switching** and **quantile regression neural network (QRNN)** to capture uncertainty in volatility itself—i.e., the **volatility-of-volatility**. This helps model extreme market conditions more effectively.
- This project aims to enhance **volatility forecasting** and option pricing using **neural networks**.
- These forecasts feed the **Black model** [2] to price **options on volatility**.



GARCH-LSTM & QRNN

- We build a **GARCH-LSTM hybrid model** to capture both statistical and deep-learning dynamics in financial volatility.

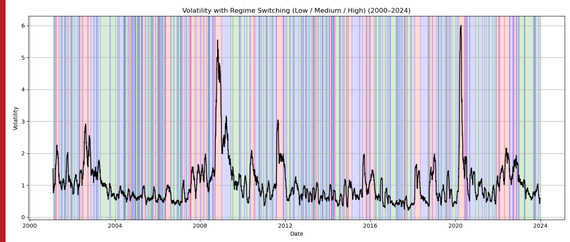


Model	MAE (Train)	RMSE (Train)	MAE (Test)	RMSE (Test)
GARCH	1.26E-03	1.79E-03	1.26E-03	1.79E-03
LSTM	3.68E-04	6.33E-04	4.09E-04	7.64E-04
LSTM-GARCH	5.74E-05	8.22E-05	7.26E-05	1.28E-04
LSTM-GARCH+VIX	8.85E-03	1.04E-02	5.92E-04	1.03E-03

- Using S&P 500 data from 2000 to 2024, we found that the **GARCH-LSTM** model provides the **best performance**.
- Unlike [1], we found that the **VIX index added little predictive power**, likely due to redundancy with GARCH, which already captures realized volatility and clustering.
- A **QRNN model** is used to model the **conditional distribution of volatility** by predicting quantiles, rather than just the mean.
- The **QRNN minimizes the pinball loss**, allowing the model to evaluate forecast accuracy at different levels of market risk.
- **Lower quantiles are generally easier to predict**, as volatility tends to be more stable during calm market periods, whereas higher quantiles capture extreme events and are more challenging.

Regime-Switching

- To better capture structural shifts in market volatility, we introduce a **Regime-Switching** framework based on rolling volatility thresholds.
- The model uses a **rolling window of historical 1-year volatility** to compute a threshold (25th, 75th percentile), defining **three regimes** (high in red, medium in blue, low in green).



Regime	Count (%)	MAE	RMSE
Low Volatility	1806 (31%)	2.88E-04	5.33E-04
Medium Volatility	2634 (46%)	4.15E-04	7.21E-04
High Volatility	1324 (23%)	1.02E-03	1.52E-03

- **Best accuracy** was achieved in **low-volatility periods**, followed by medium-volatility. High-volatility regimes had the largest errors, showing the model's **difficulty in capturing sudden shifts** during stressed markets.
- **Forecast errors increased** with regime volatility, highlighting the model's **reduced accuracy** under **high-stress conditions**.
- Overall, **regime switching** helps separate market states and **reveals model strengths and weaknesses**, guiding improvements during volatile periods.

Volatility Option Pricing

- Our **LSTM–GARCH** model generates **forward-looking volatility forecasts** that are fed into the **Black (1976) model** to compute the fair value of volatility options.
- This framework ensures **consistent, no-arbitrage pricing** of calls and puts across different strikes and maturities, allowing for robust comparison and strategy design.

K/T	1 month	3 months	6 months
23.672	8.2604	11.2846	14.0927
26.631	6.6299	10.0343	13.1018
32.549	4.1969	7.9923	11.4184
35.508	3.3215	7.1592	10.6989

- The table shows **4 strikes (K) × 3 maturities (T)**, illustrating how **option prices decrease as strike prices rise** but **increase** with **longer maturities** due to greater uncertainty.
- This pricing surface highlights how **model-implied volatilities** can directly translate into actionable option valuations for **hedging** or **volatility trading** strategies.
- Overall, the integration of LSTM–GARCH with the Black model provides a **data-driven, adaptable tool** for pricing under evolving market conditions.

References

[1] Black, F. (1976). The pricing of commodity contracts, *Journal of Financial Economics*, 3, 167-179.

[2] Roszyk, N., and Ślepaczuk, R. (2024). The hybrid forecast of S&P 500 volatility ensembled from VIX, GARCH and LSTM models, arXiv:2407.16780v1.